

```
from google.colab import files
uploaded = files.upload()
```

Screenshot... 142745.png

Screenshot 2026-08-07 142745.png(image/png) - 94094 bytes, last modified: 8/7/2026 - 100% done
Saving Screenshot 2026-08-07 142745.png to Screenshot 2026-08-07 142745.png

◆ Gemini

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# -----
# Load the image
# -----
# Fix: Changed the image path to use one of the uploaded file names.
# Please adjust "Screenshot 2026-08-07 142745.png" if you intend to use a different upload
image = cv2.imread("Screenshot 2026-08-07 142745.png")

if image is None:
    print("Error: Image not found! Please check the file name and ensure it's uploaded.")
    exit()

image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# -----
# Source points (Original Image)
# -----
src_pts = np.float32([
    [50, 50],
    [300, 50],
    [300, 300],
    [50, 300]
])
```

```
# -----  
# Destination points (Transformed Image)  
# -----  
dst_pts = np.float32([  
    [10, 100],  
    [350, 50],  
    [300, 350],  
    [80, 300]  
])  
  
# -----  
# Compute Homography using DLT  
# -----  
H, status = cv2.findHomography(src_pts, dst_pts, method=0)  
  
print("Homography Matrix (DLT):")  
print(H)  
  
# -----  
# Apply Perspective Transformation  
# -----  
height, width = image.shape[:2]  
transformed = cv2.warpPerspective(image_rgb, H, (width, height))  
  
# -----  
# Display Results  
# -----  
plt.figure(figsize=(10,5))  
  
plt.subplot(1,2,1)  
plt.imshow(image_rgb)  
plt.title("Original Image")  
plt.axis("off")  
  
plt.subplot(1,2,2)  
plt.imshow(transformed)  
plt.title("DLT Transformed Image")  
plt.axis("off")
```

```
plt.show()
```

Homography Matrix (DLT):

```
[[ 7.94698085e-01  4.28571429e-01 -5.10751105e+01]  
 [-2.84241532e-01  1.35493373e+00  4.73490427e+01]  
 [-1.64948454e-03  1.82621502e-03  1.00000000e+00]]
```

Original Image



DLT Transformed Image



